PionCloud development roadmap

models, analyses and publications that can extend the calculator

PionCloud maintainers (TDIS)

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PionCloud computes tagged structure functions as

$$F_2^{(MB)}(x;\text{cuts}) = \int_{\text{region}} dy\, dk_T^2\ \underbrace{f_{MB}(y,k_T)}_{\text{splitting}}\ \underbrace{F_2^M(x/y,Q^2)}_{\text{meson SF}}, \tag{1}$$

and since v1.4 the two physics ingredients are interfaces (`SplittingModel`, `PionSFModel` in `include/Models.hh`). Most items below are therefore *new classes*, not changes to the integration code. References marked [to verify] were identified but not yet read in full.

Priority list

	item	where in code	effort	needs
1	Chiral-EFT on-shell πN, $\pi\Delta$ splitting functions	`SplittingModel`	small	— (formulas in `docs/notes/piDelta_check.cc`)
2	Modern pion PDFs (JAM, xFitter) + target-mass corrections	`PionSFModel`	medium	table reader (or LHAPDF: on hold)
3	t-scan, x_π output; cross sections and rates	scan modes, output	medium	$R = \sigma_L/\sigma_T$, luminosity, acceptance
4	HERA leading-neutron validation test	`tests/`	medium	H1/ZEUS tables
5	Kaon cloud ($K\Lambda$, $K\Sigma$, $K\Sigma^*$) + kaon PDFs	both interfaces	large	nonlocal SU(3) EFT formulas
6	Deuteron target (TDIS-n); absorption and off-shell systematics	new module	large	spectator-tagging theory

1 Meson structure function (`PionSFModel`)

Why. GRV92 (1992, LO) and GRS99 are the only sets. The pion PDFs have since been determined in global fits, including the Sullivan-process data that TDIS itself will extend. The corrected GRS99 already differs from GRV92 by 5–17% on the TDIS tagged curves: the PDF choice is a real model uncertainty of any F_2^π extraction.

Sources.

- JAM: P. C. Barry *et al.*, PRL **121**, 152001 (2018) (Drell–Yan + HERA leading neutron); P. C. Barry *et al.*, PRL **127**, 232001 (2021), arXiv:2108.05822 (threshold resummation, large-x behaviour $(1-x)^1$ vs. $(1-x)^2$); N. Y. Cao *et al.*, PRD **103**, 114014 (2021) (q_T-dependent Drell–Yan); simultaneous pion+kaon fit with lattice-QCD moments, PRD (2025), doi:10.1103/ny6w-1n19 (kaon $\bar{u}$ large-x exponent 1.6 vs. 1.16 for the pion).
- xFitter pion PDFs: I. Novikov *et al.*, PRD **102**, 014040 (2020).

- Continuum Schwinger-function (DSE) pion and kaon PDFs: Z.-F. Cui, C. D. Roberts *et al.*, 2020–2022 [to verify] (parameter-free predictions; contrasting model).

Implementation. (a) `TablePionSF`: read published $xf(x, Q^2)$ grids (LHAPDF `.dat` format is plain text) and interpolate in $(\log x, \log Q^2)$; no external library. Use replicas/eigensets for a PDF band. (b) LHAPDF backend later (on hold). (c) Target-mass corrections and NLO: at $Q^2 \approx 1$–3 GeV^2 and $x_\pi \lesssim 0.8$ these are not small; review: I. Schienbein *et al.*, J. Phys. G **35**, 053101 (2008). Implement as an option of the `PionSFModel` (the LO relation $F_2 = \sum e_q^2 x(q + \bar{q})$ stays the Fortran-compatible default).

Test. Sum rules (as for GRS99); reproduce published curves of the JAM pion F_2 at one Q^2.

2 Splitting functions (`SplittingModel`)

1. Chiral-EFT on-shell forms (J. R. McKenney *et al.*, PRD **93**, 054011 (2016), Eqs. 2 and 5). The $\pi\Delta$ integrand is already implemented and checked ($\langle n \rangle_{\pi\Delta} = 0.058$ vs. 0.06 published; `docs/notes/piDelta_normalisation.pdf`). Adding `PiN_ChiralEFT` and `PiDelta_ChiralEFT` gives a second model alongside the Hobbs/HSS (Sullivan-form) one. *Caveat:* the cut-offs must be those fitted with these forms (their Table I), not HSS values. *Test:* $\langle n \rangle$ of Table I for all regulators.

2. Nonlocal chiral SU(3) effective theory (Y. Salamu, C.-R. Ji, W. Melnitchouk, A. W. Thomas, P. Wang): PRD **99**, 014041 (2019), arXiv:1806.07551 (splitting functions); PRD **100**, 094026 (2019) (flavour asymmetries); review arXiv:2509.25657 (2025). Octet and decuplet intermediate states (πN, $\pi\Delta$, $K\Lambda$, $K\Sigma$, $K\Sigma^*$) with on-shell, off-shell, δ-function and nonlocal terms. This is the natural source of the **kaon cloud** for the kaon TDIS run (E12-15-006A, Λ/Σ tagging) and the current standard for the $\bar{d} - \bar{u}$ connection. *Work:* new channels (masses, SU(3) couplings), kaon structure function (item 1 sources), $\Lambda \to p\pi^-$ tagging kinematics.

3. Regge exchange and absorption. Bishari-type Reggeized pion (McKenney *et al.*, Eq. 13); Reggeized exchange with absorptive (rescattering) corrections by Kopeliovich *et al.* [to verify]. Absorption reduces the tagged rate; its size at JLab energies should be quantified as a systematic.

4. Off-shell pion structure. DSE studies find the meson behaves as on-shell up to $|t| \approx 0.6$ GeV^2 (pion) and 0.9 GeV^2 (kaon) (A. C. Aguilar *et al.*, EPJA **55**, 190 (2019), arXiv:1907.08218, and references therein). Implement as a t-dependent validity warning or an optional off-shell modification factor.

3 Observables and kinematics

- **t-scan and x_π output**: F_2 in bins of $t = -(k_T^2 + y^2 M^2)/(1 - y)$ at fixed x (proposal PR12-15-006 Fig. 36), and $x_\pi = x/y$. The kbin kernel (`kbinF2Exact`) already integrates in $(y, u = k_T^2)$; t bins are a change of the integration region.
- **Cross sections and rates**: $d\sigma/dx\,dQ^2$ from F_2 and $R = \sigma_L/\sigma_T$, times luminosity, acceptance (mTPC, SBS) and beam time ⇒ counts per $(x, |\vec{k}|, t)$ bin; this is what the TDIS projections need. Generator interface: `eval()` already exists.
- **$|\vec{k}|$-scan** (proposal Fig. 45).
- **Nucleon reference** F_2: replace CTEQ6 (2002) by CJ15 (A. Accardi *et al.*, PRD **93**, 114017 (2016)), which treats large x and the neutron/deuteron carefully; may also explain the 3–7% offset of the proposal's $F_2^{p,n}$ curves.

- **Deuteron target (TDIS-n)**: Fermi motion, final-state interactions, nucleon off-shellness in spectator-tagged DIS (W. Cosyn and C. Weiss; M. Strikman and C. Weiss) [to verify].

4 Validation and the wider landscape

- **HERA leading neutrons** (H1, EPJC **68**, 381 (2010); ZEUS): McKenney *et al.* fit them with a t-exponential, $\Lambda = 0.927$ GeV. A second "published-curve" ctest, like `PionCloudValidation`. Recent benchmark: *Leading-neutron electroproduction at HERA and the EIC: Sullivan process, target fragmentation, and pion PDFs*, arXiv:2609.03368 (2026) [to verify]: Sullivan dominance mainly for $x_L \gtrsim 0.6$, JAM21 replicas with five flux regulators.
- **EIC / EicC**: Feasibility study of pion and kaon structure via the Sullivan process at EicC, arXiv:2512.01720 (projected statistical uncertainties $< 5\%$ pion, $< 8\%$ kaon); kaon structure at EicC, Chin. Phys. C (2022); ECCE detector study, arXiv:2208.14575; Aguilar *et al.* (2019) above. Small-x kinematics and leading-neutron variables (x_L, p_T) would make PionCloud usable as the model back-end of these studies.

5 Start-here checklist for the next session

1. `git fetch`; work from branch `v1.4-improvements` (not yet merged into `main`; decide whether to open the PR first).
2. Build, `ctest` (2 tests: 216 unit + 28 validation checks must pass).
3. Item 1: add `PiN_ChiralEFT` / `PiDelta_ChiralEFT` as `SplittingModels` (formulas from `docs/notes/piDelta_che` register them (`splitting_model = piN_chiralEFT`), test $\langle n \rangle$ against McKenney Table I, document in `CHANGELOG.md`.
4. Follow `CONTRIBUTING.md`: bit-identical defaults, or a reference, a measured effect and a failing-then-passing test.